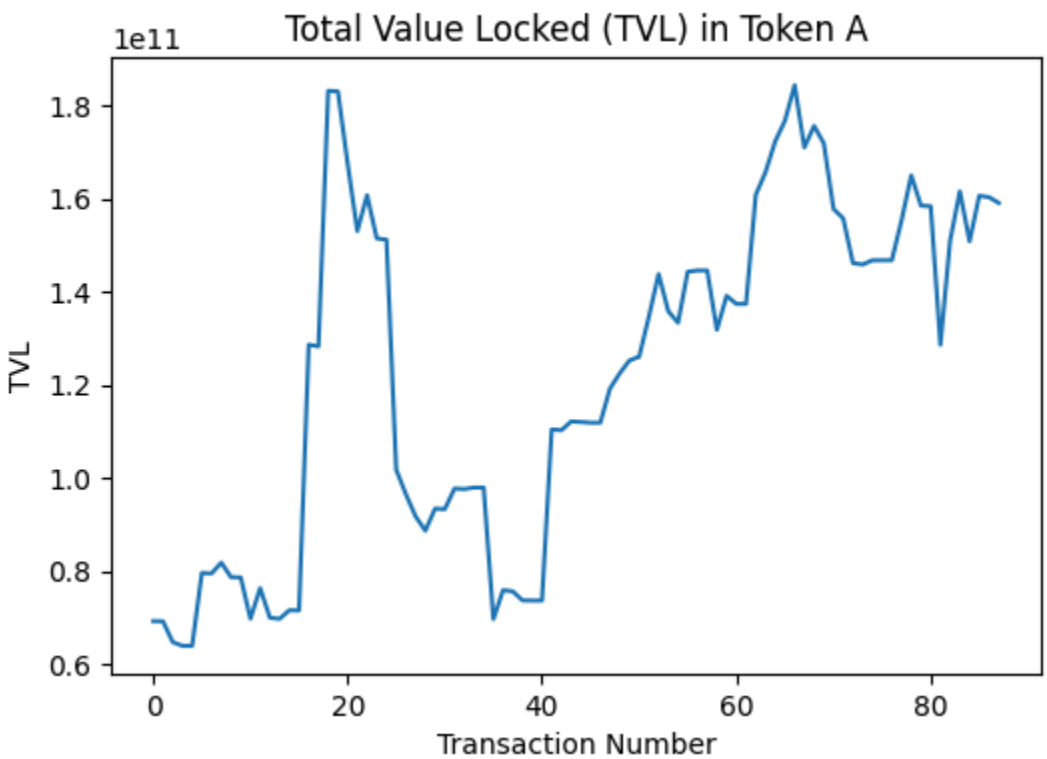
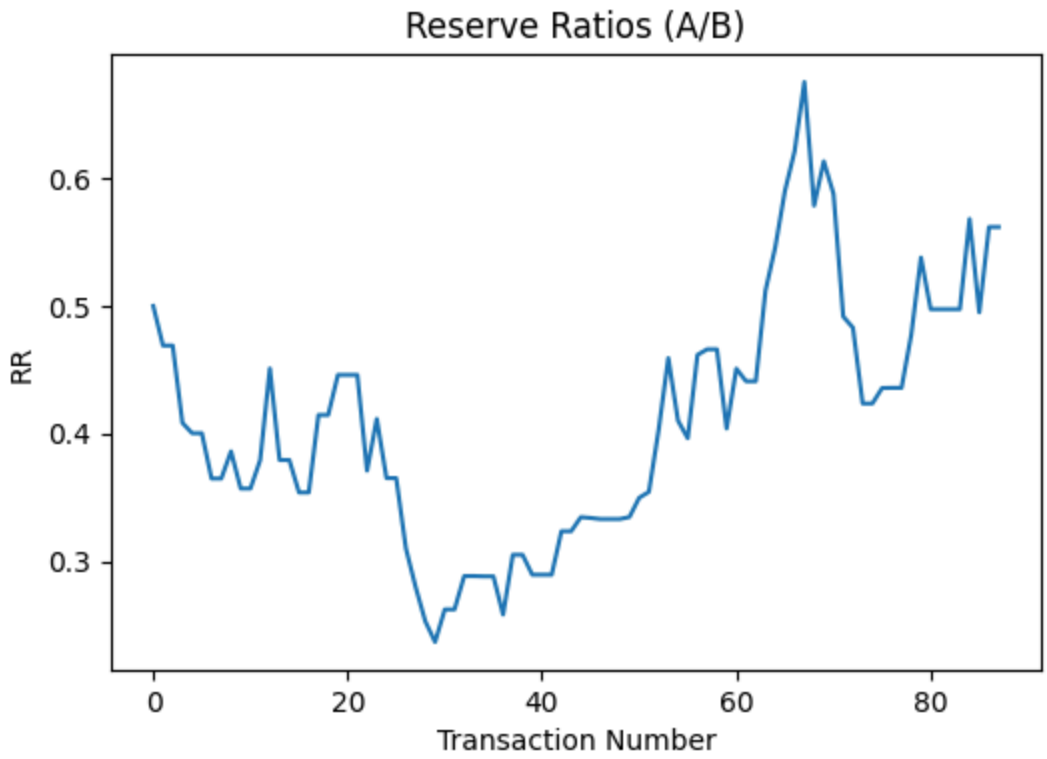
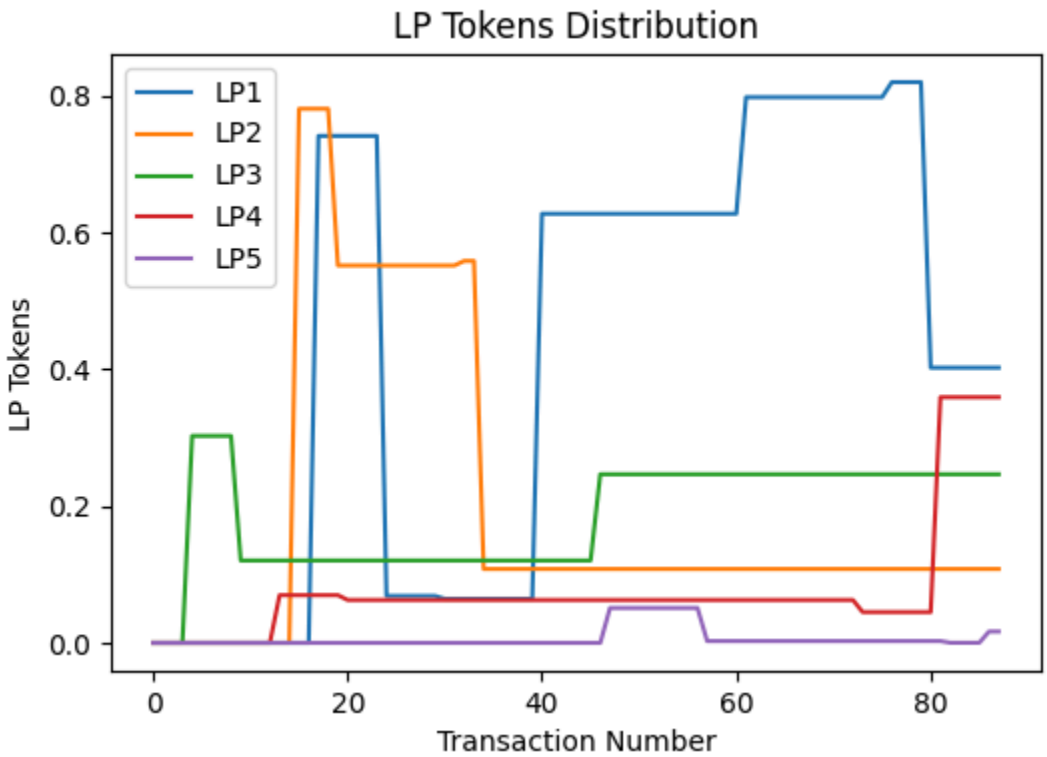
Security and Validations

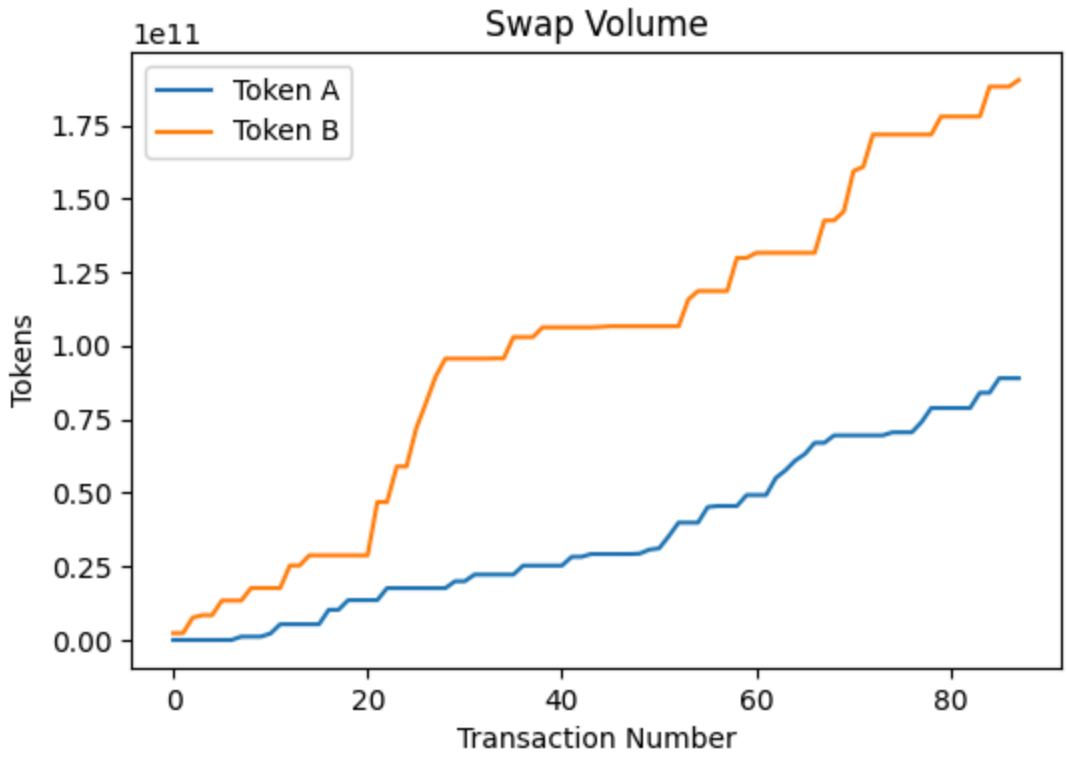
1. To combat re-entrancy attacks:
    a. We burn/transfer tokens away from msg.sender first, only then modify any state variables.
    b. We transfer any tokens to the contract caller only after modifying any state variables.
2. To mitigate integer division issues:
    a. When we want a standalone ratio, we represent it by multiplying it with a big integer (10^18).
    b. When we are multiplying fractions together, we first multiply all numerators and then divide by all denominators.
3. LP Tokens can only be minted or burnt by the DEX. The DEX can only be set once in the LP Token contract and that too only by the owner who deployed the LP Token contract.
4. Swaps are not allowed when reserves haven't been set up.
5. Deposits are reverted if they don't approximately maintain the reserve ratio.
6. All contract calls are reverted if any transfers fail.

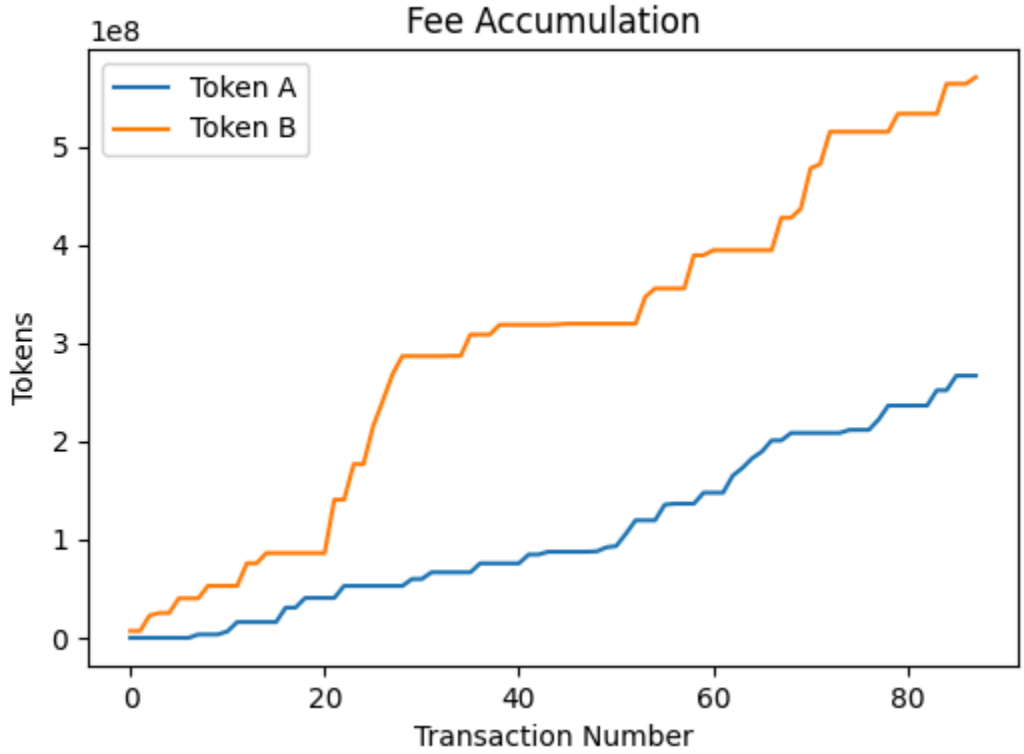
Metrics

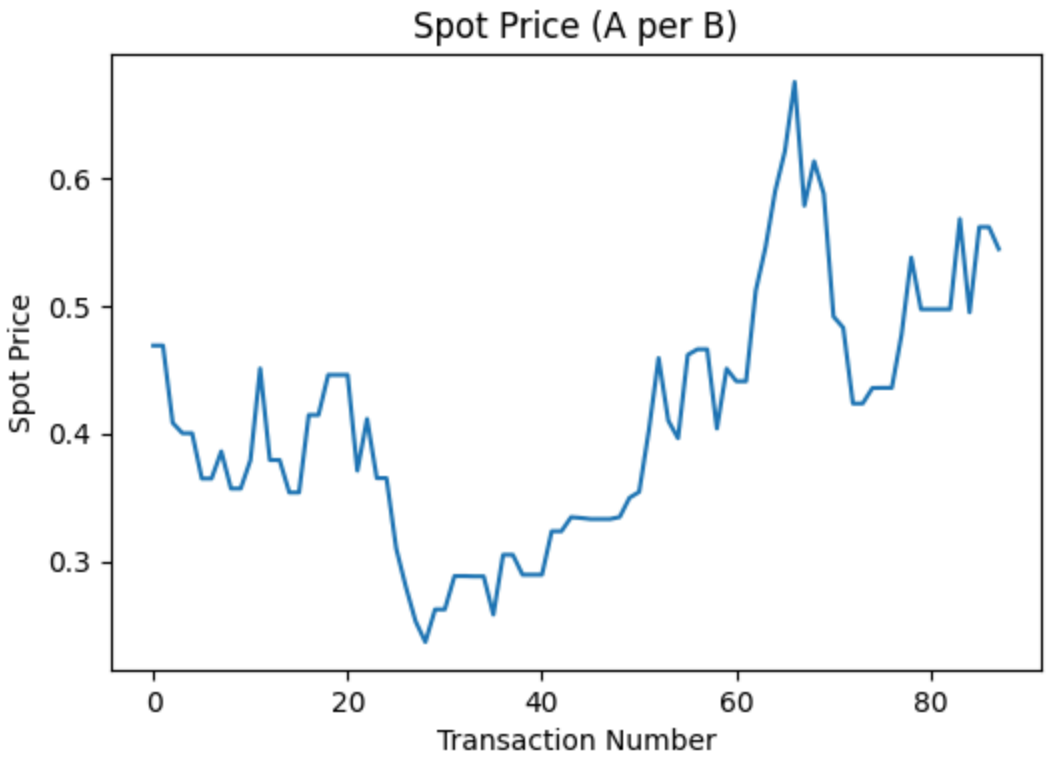


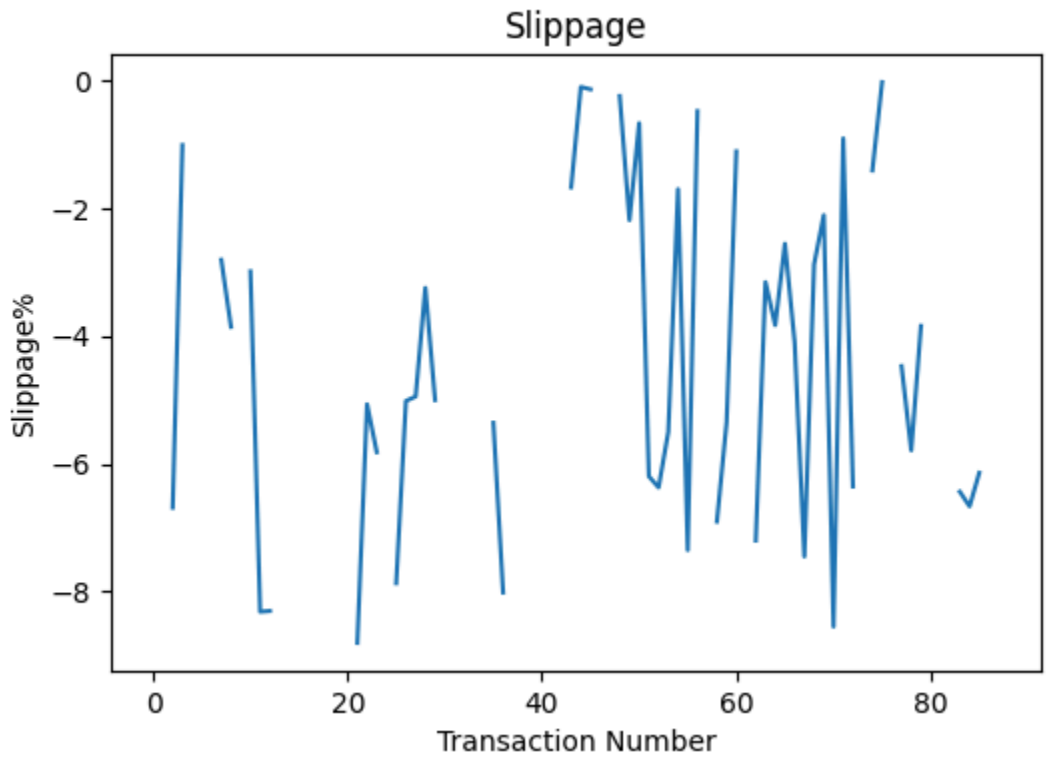

Reserve Ratios (A/B)
RR
0.6
0.5
0.4
0.3
0
20
40
60
80
Transaction Number


LP Tokens Distribution
LP1
LP2
LP3
LP4
LP5
LP Tokens
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
Transaction Number


Swap Volume
1e11
Token A
Token B
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Tokens
0
20
40
60
80
Transaction Number


Fee Accumulation
1e8
Token A
Token B
5
4
3
2
1
0
Tokens
0
20
40
60
80
Transaction Number


Spot Price (A per B)
Spot Price
0.6
0.5
0.4
0.3
0
20
40
60
80
Transaction Number


Slippage
Slippage%
0
−2
−4
−6
−8
0
20
40
60
80
Transaction Number

Questions

1. Only the corresponding DEX should be allowed to mint/burn LP tokens.
2. DEXes provide retail investors with an open market that gives them a fair way to trade among tokens.
3. Based on the transactions, the miner will be able to predict when the prices of certain tokens will go up or down and by how much. Using this information, miners can be sure to make a profit by putting their own trades, buying low and selling high. This can be mitigated by
    a. Ensuring the miner's mempool is public
    b. Batching transactions into a single transaction
    c. Large liquidity pool
4. Gas fees are paid by the traders who want to swap or by LPs who want to deposit or withdraw. Gas fees will be largely the same regardless of amount, so it won't make sense to do small trades after a certain point. The same applies to arbitrage.
5. The one who has more gas gets the freedom to transact more often, which means they will be able to grab more opportunities for arbitrage. It would also give them more freedom to do more trades with lesser volume, which would reduce their risk.
6. The best way to minimize slippage is to have a high amount of liquidity in the DEX. Other possible mitigation could be slippage limits on swaps.
7. (S+1) * TLF = Y received / Y reserves

$$TLF = \frac{X_{des}}{X_{res}}$$

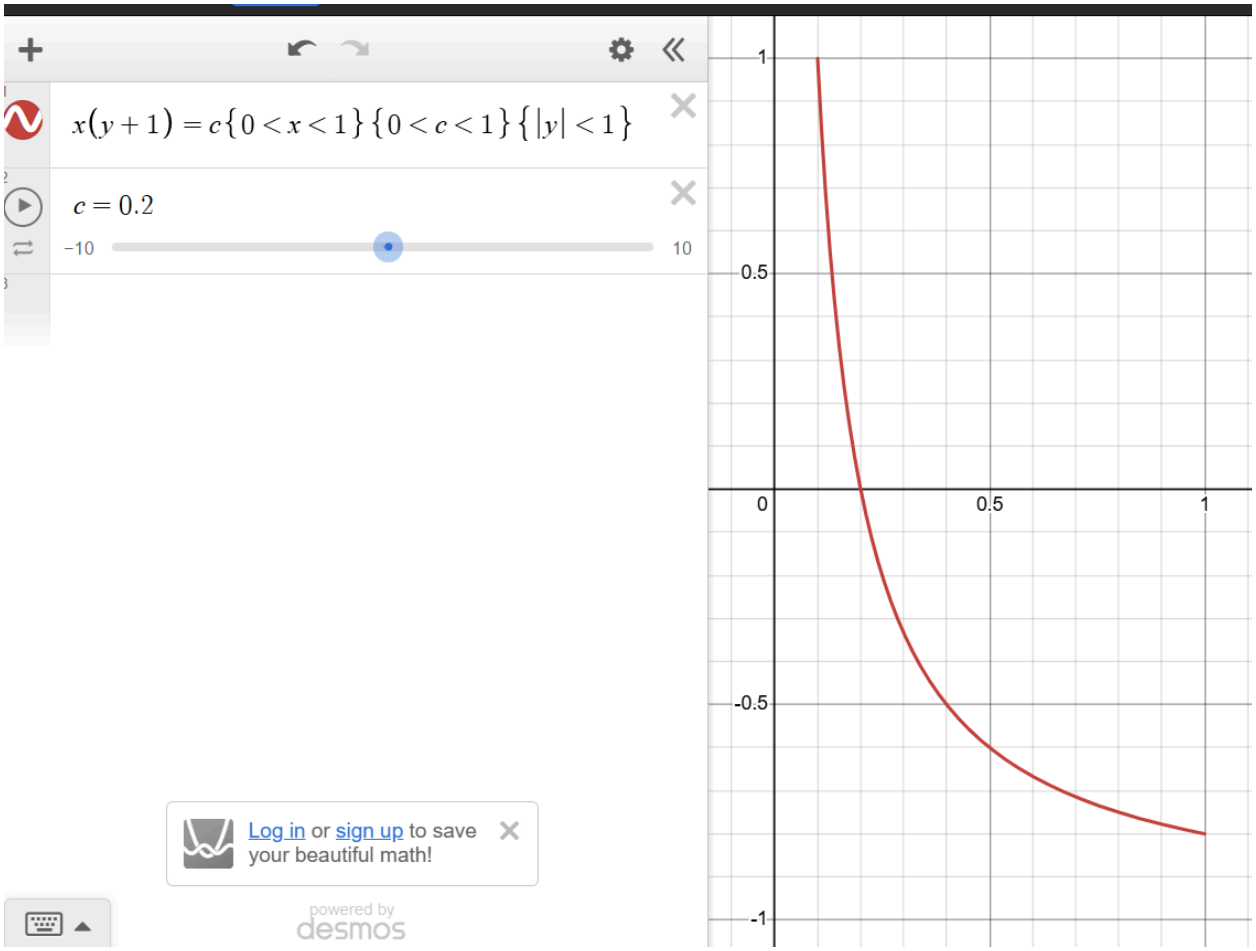
$$S = \frac{\frac{Y_{rec}}{X_{des}} - \frac{Y_{res}}{X_{res}}}{\frac{Y_{res}}{X_{res}}}$$

$$= \frac{\frac{X_{res}}{X_{des}} Y_{rec} - Y_{res}}{Y_{res}}$$

$$S = \frac{Y_{rec}}{TLF\, Y_{res}} - 1$$

$$\Rightarrow (S+1)\, TLF = \frac{Y_{rec}}{Y_{res}}$$

This will be a hyperbola, with S centered on -1, TLF=0



TLF is x axis, Slippage is y. c = Y received / Y reserves

We see that there is zero slippage when TLF = X swapped / X reserves = Y received / Y reserves. If the mismatch is too large, slippage occurs.